Omar Hafez Khalil

Artificial Intelligence Engineer

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Objective

AI Engineer specializing in Machine Learning, Deep Learning, and Generative AI. Experienced in building end-to-end AI systems including Computer Vision pipelines, RAG applications, and AI Agent workflows. Focused on developing production-oriented solutions using PyTorch, FastAPI, and modern LLM frameworks.

Technical Skills

Deep Learning & Machine Learning: Neural Networks | Computer Vision | Object Detection | Transfer Learning | Fine-Tuning | Hyperparameter Tuning | Model Evaluation | Feature Engineering | Image Processing | Data Processing | Data Visualization

Frameworks & Libraries: Python | PyTorch | scikit-learn | Ultralytics | LangChain | LangGraph | Transformers | OpenCV | Pandas | NumPy

Generative AI & LLM Engineering: Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | AI Agents | Vector Databases | ChromaDB | Embeddings | Prompt Engineering | Local LLM Deployment (Ollama)

Backend & Databases: FastAPI | REST APIs | Node.js | Express.js | MongoDB | Mongoose | SQL | Database Design

Developer Tools: Git | GitHub | Postman | Jupyter Notebook | Google Colab

Soft Skills: Problem Solving | Communication | Teamwork | Time Management | Continuous Learning

Projects

Venezia Bank – AI-Powered Digital Banking System | *AI, Flutter, Laravel, MySQL, FastAPI*

- Developed a comprehensive AI-powered digital banking ecosystem featuring a cross-platform mobile application for customers and a web-based platform for bank employees, supported by a Laravel backend and integrated AI services. The platform provides core banking services including payments, transfers, account management, loans, investments, financial reporting, and fraud monitoring.
- Served as an AI Engineer responsible for designing and implementing 4 core AI modules, contributing to an ecosystem integrating 8+ AI capabilities, including financial fraud detection, receipt recognition, customer churn prediction, product recommendation, stock prediction, loan decision support, employee decision support, and an LLM/RAG financial chatbot.
- Designed and deployed AI services as FastAPI-based microservices, enabling integration between AI models and the core banking platform through RESTful APIs.
- GitHub: https://github.com/abdulaziz-bawabeh/venezia-bank-ai-system

Receipt Recognition System | *FastAPI, Ultralytics(YOLOv8), PyTorch, ResNet-50, Gemini API, OpenCV*

- Developed an end-to-end AI system that transforms receipt images into structured financial data through a complete pipeline including image validation, object detection, OCR, and product classification. Built and deployed the system as a unified FastAPI service for automated financial document processing.
- Image Validation Accuracy: 98.83%
- Region Detection mAP50: 0.7693
- Product Classification F1-Score: 0.93
- GitHub: https://github.com/OmarHKhalil/Receipt-Recognition-System

DocMind AI (Multi-Agent) | *Python, LangChain, LangGraph, ChromaDB, Ollama, Streamlit*

- Built an intelligent multi-agent document analysis platform that enables users to interact with documents through specialized AI agents. Implemented hybrid retrieval with local LLM inference to deliver accurate, context-aware responses while preserving data privacy.
- GitHub: https://github.com/OmarHKhalil/AI-Agents-Projects/tree/main/DocMind_AI_Multi-Agent

Banking Finance Chatbot (RAG) | *FastAPI, LangChain, ChromaDB, Ollama, Gemini API, Embeddings*

- Designed and developed a production-oriented Retrieval-Augmented Generation (RAG) chatbot for banking and financial assistance. Integrated vector search with large language models to generate accurate, domain-specific responses while maintaining a professional banking assistant persona.

- BERTScore: 88.29%
- GitHub: https://github.com/OmarHKhalil/LLM-RAG-Projects/tree/main/Genai-Finance-Chatbot

AI-powered Banking Retention & Product Recommendation System | *FastAPI, XGBoost, LightGBM*

- Developed an intelligent banking retention system that predicts customer churn and automatically recommends personalized financial products through a real-time REST API. Implemented a dual-machine-learning pipeline for prediction and recommendation.
- Churn Prediction F1-Score: 0.88
- Product Recommendation F1-Score: 0.88
- GitHub: https://github.com/OmarHKhalil/Bank-Retention-Recommendation
- Demo: https://huggingface.co/spaces/Omarsy2/Bank-Retention-Recommendation

NutriVision: Food Classification | *PyTorch, ResNet50, EfficientNet-B0, TensorBoard, TorchMetrics*

- Developed a deep learning system comparing custom CNN architectures with state-of-the-art transfer learning models for food image classification. Evaluated the impact of different architectures on model performance and generalization.
- 30 Food Categories
- Optimized ResNet50 architecture for resource-constrained environments, achieving high classification accuracy on 30 diverse food categories and deploying via an interactive interface.
- GitHub: https://github.com/OmarHKhalil/Deep-Learning-Projects/tree/main/NutriVision-Food-Classification
- Demo: https://huggingface.co/spaces/Omarsy2/nutrivision-app

Education

Arab International University (AIU) 2021 – 2026

- Bachelor's Degree in Informatics Engineering
- Artificial Intelligence Engineering
- GPA: 3.36/4.0

Courses

- **Agentic AI with LangChain and LangGraph** (Coursera, Completed Jun. 2026)
- **Machine Learning** (Coursera, Completed Mar. 2026)
- **Generative AI with Large Language Models** (Coursera, Completed Sep. 2025)
- **Deep Learning & Computer Vision**
  - **PyTorch for Deep Learning & Machine Learning** (freeCodeCamp, Jan. 2025 – Aug. 2025)
- **Machine Learning with Python and Scikit-Learn** (freeCodeCamp, Jul. 2024 – Dec. 2024)
- **Advanced Learning Algorithms** (Coursera, Completed Dec. 2024)
- **Unsupervised Learning, Recommenders and Reinforcement Learning** (Coursera, Completed Oct. 2024)
- **Supervised Machine Learning: Regression and Classification** (Coursera, Completed Aug. 2024)
- **Python for Data Science - Course for Beginners (Learn Python, Pandas, NumPy, Matplotlib)** (freeCodeCamp, Completed Jul. 2024)

Languages

- Arabic — Native
- English — Professional Working Proficiency